Flutter Installation Steps

A beginner-friendly setup guide for Windows 10/11

Follow these steps in order

- Install the prerequisites and download Flutter SDK
- Extract Flutter to a suitable folder
- Add Flutter to the Windows PATH
- Verify Flutter and configure Android licenses
- Connect Flutter with Android Studio
- Create and run a sample app
- Troubleshoot common setup errors

Official Flutter guide: Flutter install for Windows

Android Studio: Download Android Studio

1. Check prerequisites

• Use a supported 64-bit Windows 10 or Windows 11 computer.

• Install Android Studio and the Android SDK if you plan to build and run Android apps.

• Use a folder you can write to, such as C:\src\flutter. Avoid protected folders such as C:\Program Files.

• Ensure you have a stable internet connection and enough free disk space for the SDK, emulator and build files.

2. Download Flutter SDK

• Open https://docs.flutter.dev/get-started/install/windows.

• Follow the stable-channel download instructions and download the Windows Flutter SDK archive.

• If the page offers an automated install option that suits your setup, you may use it; the manual steps below work as a general setup approach.

3. Extract the SDK

• Create a folder such as C:\src if it does not already exist.

• Extract the Flutter archive so the SDK folder is C:\src\flutter.

• Check that this file exists: C:\src\flutter\bin\flutter.bat.

• Do not extract Flutter into a folder path with restrictive permissions.

4. Add Flutter to Windows PATH

• Press the Windows key and search for Edit environment variables for your account.

• Select the user Path variable and click Edit.

• Click New and add the Flutter bin path:

```
C:\src\flutter\bin
```

• Save all dialogs using OK.

• Close and reopen Command Prompt or PowerShell so it reads the updated PATH.

5. Verify the installation

Open Command Prompt or PowerShell and run:

```
flutter --version
flutter doctor
```

The version command should print the installed Flutter version. Flutter doctor checks the development setup and lists anything that needs attention.

6. Configure Android development

• Open Android Studio → More Actions → SDK Manager (or Tools → SDK Manager inside a project).

• In SDK Tools, install Android SDK Platform-Tools, Android SDK Command-line Tools (latest), Android SDK Build-Tools and Android Emulator if needed.

• Open a terminal and run:

```
flutter doctor --android-licenses
```

Review and accept the license prompts. Then run flutter doctor again.

7. Install Flutter and Dart plugins

• Open Android Studio → Settings → Plugins.

- Search for Flutter and select Install. Install Dart if prompted.
- Restart Android Studio.
- These plugins provide IDE support; the Flutter SDK still needs to be installed separately and available on PATH.

8. Create your first Flutter app

In Command Prompt or the Android Studio terminal, run:

```
flutter create my_first_app
cd my_first_app
flutter run
```

• Start an Android emulator from Android Studio Device Manager, or connect an Android phone with Developer options and USB debugging enabled.

• Run flutter devices to check whether Flutter detects your device.

• Run flutter run from the project folder. The first build may take several minutes while dependencies download.

9. Open the project in Android Studio

• Choose Open in Android Studio.

• Select the my_first_app folder containing pubspec.yaml.

• Wait for Gradle and Dart/Flutter project setup to finish.

• Select a device in the toolbar and click Run.

10. Common errors and fixes

Issue	Suggested fix
'flutter' is not recognized	Check that the correct Flutter bin folder is in PATH, then reopen the terminal.
Android licenses not accepted	Run flutter doctor --android-licenses and accept the prompts.
No connected devices	Start the emulator or connect a phone; run flutter devices.
Android toolchain error	Open SDK Manager and install the SDK tools listed by flutter doctor.
Emulator fails to start	Check virtualization support; try a supported system image or a physical device.
pub get network error	Check internet/proxy settings and retry flutter pub get.

11. Final checklist

■ Flutter SDK extracted correctly.

■ Flutter bin folder added to PATH.

■ flutter --version works.

■ flutter doctor reviewed and Android licenses accepted.

■ Android Studio Flutter/Dart plugins installed.

■ Emulator or phone detected.

■ Sample app successfully runs.

Note: Folder paths and Android Studio menu labels can vary. Follow the official Flutter guide if the current download page offers a different installation flow.